Sea Ice Classification — Project Summary

A plain-language summary of the three models we built to label each pixel of a satellite image as **thick ice**, **thin ice**, or **water**. We trained on two Antarctic tiles (T02CNA, T02CNC) and tested on a third never-seen tile (T03CWT). The scores below are all measured on that held-out test tile.

Two kinds of data are available for every region: a Sentinel-2 RGB satellite image (looks down from above, picks up color/texture), and a set of ICESat-2 photon measurements (laser altimeter, gives precise heights along narrow ground tracks). The interesting question is *can we use both together to do better than either alone?*

Headline results (test tile T03CWT)

Model	Test mIoU
U-Net (image only)	0.8704
LSTM (photon CSV only) — sweep winner	0.6978
Deep Fusion (image + LSTM) — OUR BEST	0.9010

Higher is better; 1.0 would be perfect. The image-only U-Net already does well by itself (0.87) — the satellite picture is informative. The photon-only LSTM does much worse alone (0.70) because each photon track only covers a thin line across the patch. But **combining them (0.90) beats both**, especially on the rare thin-ice and water classes.

1. U-Net — image only

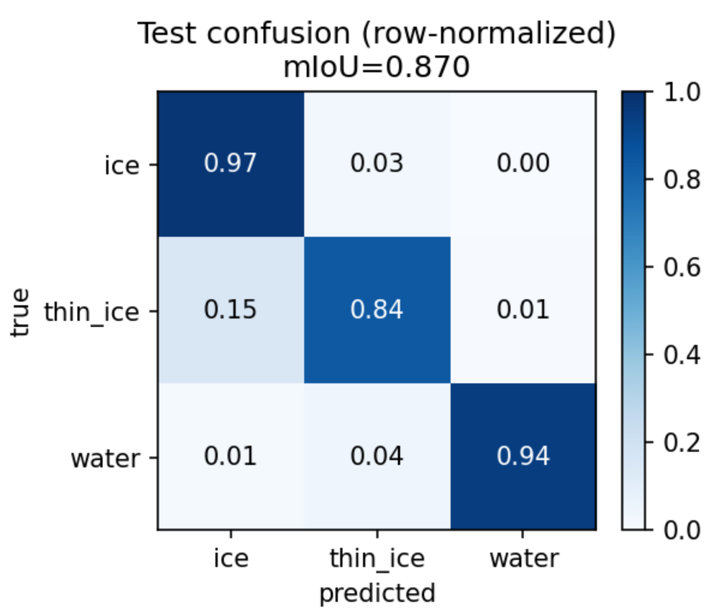
U-Net is a standard image-segmentation neural network. It takes the RGB satellite picture as input and produces a 3-class label for every pixel. We used a ResNet-18 encoder pre-trained on ImageNet, which gives the model a head-start: it already knows how to recognize edges, textures, and shapes before it even sees sea ice.

U-Net is good at things the image makes obvious — large patches of open water look very different from packed ice. Where it struggles is thin ice, which can look very similar to thick ice in the visible spectrum.

Benchmarks

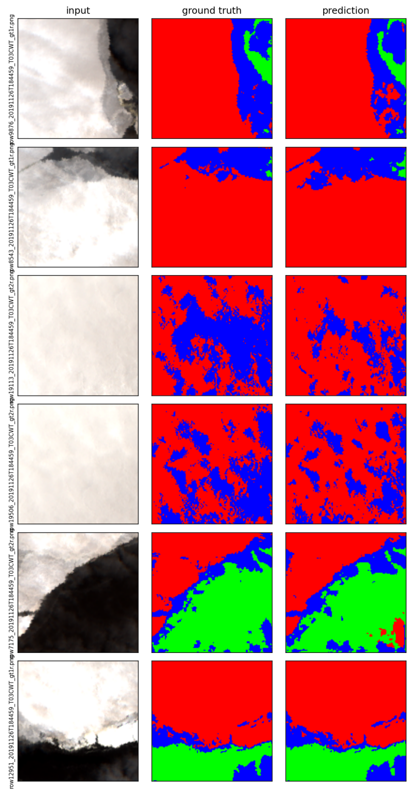
Metric	Value
Test mIoU (average IoU across the 3 classes)	0.8704
Pixel accuracy	0.9429
IoU — thick ice	0.9299
IoU — thin ice	0.7683
IoU — water	0.9130

Confusion matrix — U-Net



Each row shows what percentage of one true class got predicted as thick ice, thin ice, or water. A perfect model would have 100% on the diagonal.

Sample predictions — U-Net



Left column: the satellite image the model saw. Middle: the human-drawn ground truth. Right: what U-Net predicted. Red = thick ice, blue = thin ice, green = water.

2. LSTM — photon CSV only

LSTM stands for “Long Short-Term Memory”. It is a neural network that reads a *sequence* of measurements one at a time and remembers context as it goes. Perfect for time-series or, in our case, a series of 10-meter segments along a ICESat-2 ground track.

We followed the professor’s own LSTM recipe (her Keras notebook for the same problem):

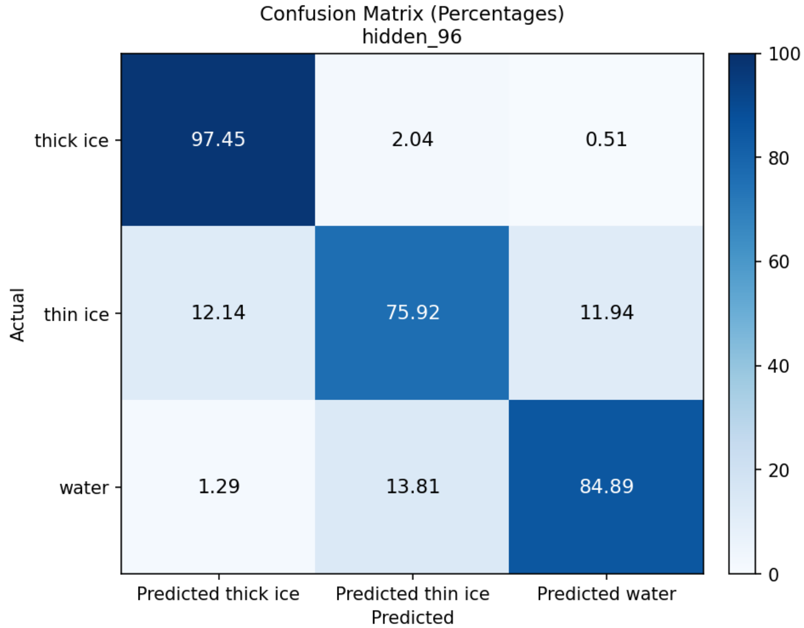
Setting	Value
LSTM direction	uni-directional (forwards only)
Hidden units	96
Number of layers	1
Input window	5 segments (center ± 2)
Features per segment	8 engineered numbers
Loss function	Focal loss, α=[0.05, 0.45, 0.60], γ=2.0
Optimizer	Adam, learning rate 8.886e-4

We didn’t guess these numbers — we ran a 21-config sweep that varied the alpha vector, gamma, hidden size, learning rate, dropout, window length and random seed one axis at a time. The winner was **hidden = 96** — everything else stayed close to the professor’s defaults. The 8 features are physically meaningful: mean / median / standard-deviation of corrected photon heights, photon counts, background counts, background rate, and two derived height-asymmetry features.

Benchmarks

Metric	Value
Test mIoU	0.6978
Pixel accuracy	0.9594
IoU — thick ice	0.9671
IoU — thin ice	0.5427
IoU — water	0.5836
Macro F1	0.8080

Confusion matrix — LSTM



Strong diagonal on thick ice (97%) because that's the most common class. Thin ice (76%) and water (85%) are weaker because the LSTM only sees a few photon segments per patch — it has no idea what the surrounding 128×128 image actually looks like.

3. Deep Fusion — image + LSTM — the winner

This is the model that won. It uses both data sources, and it lets the two networks compare notes inside the architecture (not just by averaging their predictions at the end).

How it works:

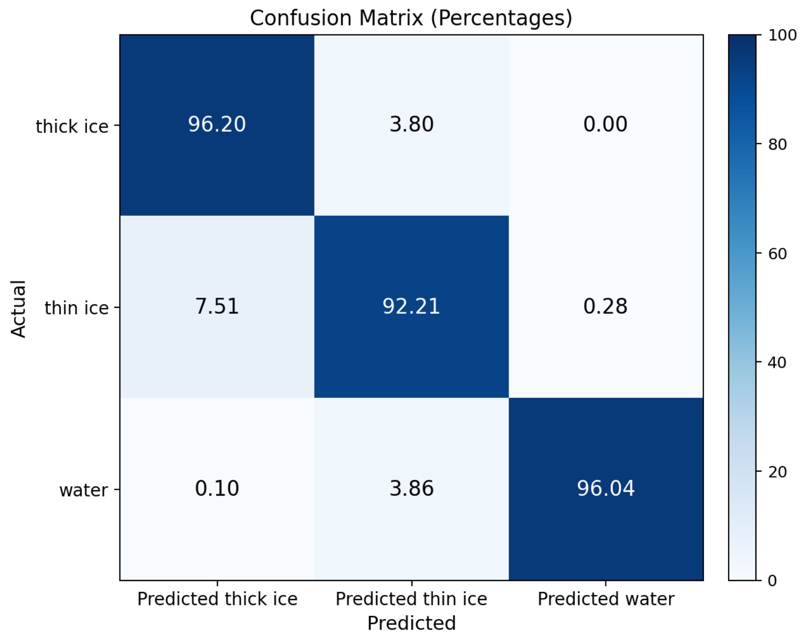
• The image branch is the same U-Net from Model 1, which turns the satellite picture into a feature map.
• The CSV branch is the **LSTM winner from Model 2**, loaded with the weights it learned during the sweep.
• The LSTM's output is tiled across the patch and concatenated with the U-Net's features.
• A small "attention" layer (Squeeze-and-Excitation) learns to rescale each channel of the combined features, so the model can up- or down-weight either branch depending on what it sees.
• A final 3×3 convolution outputs the per-pixel class prediction.

One trick that mattered: instead of training the LSTM from scratch inside the fusion model, we loaded the sweep-winner weights and let them adapt slowly during fusion training (10x slower than the rest of the network). This "two pretrained experts get glued together" recipe gave us +0.01 mIoU over starting from scratch.

Benchmarks

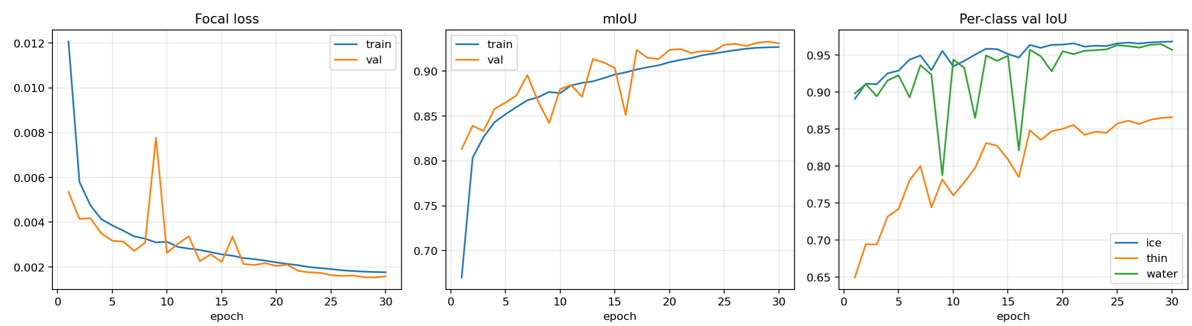
Metric	Value
Test mIoU	0.9010
Pixel accuracy	0.9530
Macro F1	0.9468
Macro precision	0.9460
Macro recall	0.9482
IoU — thick ice	0.9403
IoU — thin ice	0.8138
IoU — water	0.9489
F1 — thick ice	0.9692
F1 — thin ice	0.8973
F1 — water	0.9738

Confusion matrix — Deep Fusion (winner)



The diagonals are 96 / 92 / 96 — none below 92%. Compared to the LSTM alone (97 / 76 / 85), fusion lifted thin ice by +16 percentage points and water by +11 pp without losing thick ice.

Training curves



Left: focal loss going down on both the training and validation sets. Middle: average IoU going up. Right: per-class IoU — thin ice (the rare class) takes longer to climb but eventually stabilizes around 0.81.

Side-by-side comparison

All three models scored on the same held-out test tile (T03CWT):

Model	mIoU	Pix Acc	IoU thick ice	IoU thin ice	IoU water
U-Net (image only)	0.8704	0.9429	0.9299	0.7683	0.9130
LSTM (CSV only)	0.6978	0.9594	0.9671	0.5427	0.5836
Deep Fusion (winner)	0.9010	0.9530	0.9403	0.8138	0.9489

Key takeaways

1. The image-only U-Net is already a strong baseline (0.87 mIoU). It handles thick ice and water well but struggles with thin ice because thin ice looks visually similar to thick ice in the satellite RGB.

2. The photon-only LSTM is weak on its own (0.70 mIoU), but the photons carry real signal about ice *thickness* that the image can't see. The LSTM learns to use that.

3. When you fuse them deeply (not just averaging predictions, but combining the internal feature maps with attention), the two modalities cover each other's blind spots. Result: **0.90 mIoU**, with the biggest gains on the two minority classes (thin ice +16 pp, water +11 pp vs LSTM alone; thin ice +5 pp vs U-Net alone).

4. Loading the LSTM weights from a separate sweep and fine-tuning them slowly during fusion training was the final ingredient that pushed us past 0.90.

Everything in this PDF was measured on the held-out test tile T03CWT, which the model never saw during training. The training set was tiles T02CNA and T02CNC. Code, notebooks, and per-run metrics live under runs/ (winners) and archive/ (everything else we tried).